

COSMOS Cluster 7

Machine Learning

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Topic: Natural Language Processing (Part 2)

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Existing packages can accelerate NLP jobs

1. They define and describe a specific language in a way that makes it easy to build a ML model. The language models are built on different pools of texts, called corpus
 - a. Makes it easier to preprocess text data
2. We are going to check out two packages today
 - a. SpaCy
 - b. NLTK

Tokenization with NLTK and SpaCy

[02-Tokenization with NLTK.ipynb](#)

[03- Tokenization with SpaCy.ipynb](#)

[04-NLP Pipelines of SpaCy.ipynb](#)

SpaCy's NLP pipelines allows you to better-control text processing

[04-NLP Pipelines of SpaCy.ipynb](#)

Removing Stopwords

- Some words in a sentence do not add much information
 - “I”, “am”
- SpaCy's token has the `is_stop` attribute



is swimming difficult?



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Tools



Reddit · r/Swimming

70+ comments · 3 years ago ⋮

Why didn't you guys warn me swimming is super hard?

If your stroke is wrong it feels much more difficult. Swim a lap, stop, take a break, swim another, take a break, drink some water. Maybe try ...

[46 answers](#) · Top answer: watch effortless swimming youtube channel and keep swimming. Eit...

Why **is swimming** so hard for me? - Reddit Aug 4, 2016

A bit embarrassing, I'm 19 and can't **swim**. How hard is it to ... Jan 5, 2021

Swimming so much **harder** than running? - Reddit Jan 22, 2022

learn to **swim** as adult. impossible? - Reddit Apr 24, 2021

[More results from www.reddit.com](#)

Discuss:

How does Google know to include posts with not only the word 'swimming' but also 'swim'?

Stemming

- Stemming is the process of reducing a word to its base or root form by removing prefixes or suffixes.
- It typically uses simple rules or algorithms without considering the context or meaning of the word.
- The resulting stem may not be a valid word.

Examples of Stemming

Example:

- "running" becomes "run"
- "happily" becomes "happi"
- "better" becomes "better" (no change)

Key Things

- **Algorithmic:** Uses algorithms like Porter Stemmer, Snowball Stemmer, etc.
- **Fast:** Generally faster than lemmatization because it uses straightforward rules.
- **Less Accurate:** May produce stems that are not actual words (e.g., "ability" to "abili").
- **Use Cases:** Often used in search engines, indexing, and information retrieval.

Lemmatization

- Lemmatization is the **process of reducing a word to its base** or dictionary form, known as the **lemma**.
- Lemmatization uses the **knowledge of a language** to derive a base word
- It considers the **context** and **part of speech** to provide a **linguistically accurate root form** of the word.

Examples

- "running" becomes "run"
- "better" becomes "good"
- "children" becomes "child"

Key Things

- **Context-Aware:** Uses linguistic analysis to understand the context and part of speech.
- **Accurate:** Produces valid words as lemmas, maintaining meaning.
- **Slower:** More computationally intensive due to its complexity.
- **Use Cases:** Ideal for applications requiring linguistic accuracy, such as natural language processing, text analysis, and machine learning.

Stemming/Lemmatization with NLTK and SpaCy

[05-RemovingStopWords and Lemmatization with SpaCy.ipynb](#)

[06-Custom Lemmatization.ipynb](#)

sklearn's Vectorizer() creates bag of words

[07-Naive Bayes with sklearn.ipynb](#)